

MAACKIA

PAPILIONACEAE/LEGUMINOSAE

Genus of deciduous, summer-flowering trees or shrubs, grown for their foliage and flowers. Fully hardy. Requires full sun and fertile, well-drained soil. Propagate by seed in fall or by greenwood cuttings in early or midsummer.
M. amurensis illus. p.85

MACADAMIA

PROTEACEAE

Genus of evergreen trees, grown for their foliage and fruits. Frost tender. Prefers full light, though some shade is tolerated. Provide humus-rich, moisture-retentive but well-drained soil. Water freely while in full growth, moderately at other times. Pruning is not usually necessary, but is tolerated in fall. Propagate by seed when ripe, in fall or in spring.
M. integrifolia (Macadamia nut, Queensland nut). Spreading tree. H and S 50ft (15m). Has whorls of leathery, semi-glossy leaves and edible, brown nuts in fall. Produces panicles of small, creamy-yellow flowers in spring. Z11–12 (min. 50°F/10°C)

MACFADYENA

SYN. DOXANTHA
Cat's claw vine

BIGNONIACEAE

Genus of evergreen, woody-stemmed, tendril climbers, grown for their foxglove-like flowers. Frost hardy to frost tender. Any fertile, well-drained soil is suitable with full light. Water regularly, less when not in full growth. Provide support for stems. Thin out crowded shoots after flowering or in spring. Propagate by semiripe cuttings in summer.
M. unguis-cati, syn. *Bignonia unguis-cati*, *Doxantha unguis-cati* (Common cat's claw vine). Fast-growing, evergreen, woody-stemmed, tendril climber. H 25–30ft (8–10m). Leaves have 2 leaflets and a tendril. Bears yellow flowers, 4in (10cm) long, in late spring or early summer. Z9–11

MACKAYA

ACANTHACEAE

Genus of one species of evergreen shrub, grown for its flowers and overall appearance. Frost tender. Requires full light or partial shade and fertile, well-drained soil. Water potted plants freely when in full growth, moderately at other times. Pruning is tolerated in winter if necessary. Propagate by seed in spring or by semiripe cuttings in summer.
M. bella, syn. *Asystasia bella*. Evergreen, erect then spreading, well-branched shrub. H 3–12ft (1–4m), S 3–6ft (1–2m). Leaves are elliptic, slender-pointed, lustrous and deep green. Has spikes of narrowly funnel-shaped flowers, each with 5 large, flared, pale lilac petal lobes finely veined dark purple, from spring to fall. In warm

conditions, above 55°F (13°C), will flower into winter. Z11–12 (min. 50°F/10°C)

MACLEANIA

ERICACEAE

Genus of evergreen, spring- to summer-flowering shrubs and climbers, some scrambling, grown for their flowers. Frost tender. Requires partial shade and humus-rich, well-drained, acidic soil. Water potted specimens moderately, less when not in full growth. Shorten long shoots in winter or after flowering. Propagate by seed in spring, by semiripe cuttings in early summer, or by layering in fall.
M. insignis. Evergreen, scrambling climber with erect, sparingly branched, wandlike stems. H 10ft (3m), S 3–10ft (1–3m). Has ovate, leathery, deep green leaves, red-flushed when young. Tubular, waxy, scarlet flowers, with white tips, hang in clusters in summer. Needs support. Z11–12 (min. 10°C/50°F)

MACLEAYA

Plume poppy

PAPAVERACEAE

Genus of summer-flowering perennials, grown for their flowers and foliage. Fully hardy. Requires a position in sun and in well-drained soil. May spread rapidly. Propagate by division in late fall or early spring or by root cuttings in winter.
M. cordata, syn. *Bocconia cordata* (Plume poppy). Rhizomatous perennial. H 5ft (1.5m) or more, S 2ft (60cm) or more. Large, rounded, lobed, gray-green leaves, gray-white beneath, are produced at base of plant and up lower parts of stems. Large, feathery panicles of dainty, creamy-white flowers are borne in summer. Z3–8.
M. x kewensis 'Flamingo'. Rhizomatous perennial. H to 8ft (2.5m), S 3ft (1m). Has large, rounded, lobed, gray-green leaves, gray-white beneath. Bears large, feathery panicles with pink buds and neat, buff-pink flowers in summer. Z3–8
M. microcarpa 'Kelway's Coral Plume' illus. p.216.

MACLURA

MORACEAE

Genus of usually thorny, evergreen or deciduous trees, shrubs, and climbers, with branches often reduced to spines. Grown for their foliage and unusual fruits; both male and female trees need to be planted to obtain fruits. Fully hardy, but young plants are susceptible to frost damage. Requires full sun and needs hot summers to thrive in cold areas. Grows in any but waterlogged soil. Propagate by softwood cuttings in summer, by seed in fall, or by root cuttings in late winter.
M. aurantiaca. See *M. pomifera*.
M. pomifera, syn. *M. aurantiaca* (Osage orange). Deciduous, rounded, spreading tree. H 50ft (15m), S 40ft (12m). Has spiny shoots and ovate, dark green leaves that turn yellow in fall. Tiny, cup-shaped, yellow flowers in summer are followed on female trees by large, rounded, wrinkled, yellow-green fruits. Z5–9
M. tricuspidata, syn. *Cudrania tricuspidata*. Deciduous, compact tree. H 22ft (7m), S 20ft (6m). Bears ovate, dark

green leaves that are sometimes 3-lobed. Has small, rounded clusters of tiny, green flowers in midsummer. (borderline) Z8–10

Macrotomia echiioides. See *Arnebia pulchra*.

MACROZAMIA

ZAMIACEAE

Genus of slow-growing, evergreen shrubs and small trees, with or without trunks, grown for their palmlike appearance. Mature plants may produce conical, green flower spikes. Frost tender. Needs full light or partial shade and well-drained soil. Water containerized plants moderately when in full growth, less at other times. Propagate by seed in spring.
M. corallipes. See *M. spiralis*.
M. spiralis, syn. *M. corallipes*. Evergreen, palmlike shrub with a very short, mainly underground trunk. H and S 2–3ft (60cm–100m). Has a rosette of deep green leaves, each with a spirally twisted midrib and very narrow, leathery leaflets. Z11–12 (min. 50°F/10°C)

MAGNOLIA

SYN. MICHELIA

MAGNOLIACEAE

Genus of deciduous and evergreen trees and shrubs, grown for their showy flowers. Leaves are mainly ovate. Fully to frost hardy. Flowers and buds of early-flowering magnolias may be damaged by late frosts. Needs sun or partial shade and shelter from strong winds. Prefers fertile, well-drained soil. *M. delavayi*, *M. kobus*, *M. sieboldii* and *M. wilsonii* grow on chalky soil. Other species prefer neutral to acidic soil, but will grow in alkaline soil if deep and humus-rich. Dry, sandy soils should be generously enriched with manure and leaf mold before planting. Propagate species by semiripe cuttings in summer or by seed, when ripe, in fall, selected forms by semiripe cuttings in summer, or grafting in winter. See also feature pp.70–71.
M. acuminata (Cucumber tree). Vigorous, deciduous tree, conical when young, later spreading. H 70ft (20m), S 30ft (10m). In late spring and early summer, bears fragrant, cup-shaped, yellow-green or bluish-green flowers, to 3½in (9cm) across, amid large, ovate, pale green leaves. Flowers are followed by small, egg-shaped, red or brown fruits. Z3–8
M. 'Athene'. Vigorous, deciduous, upright to spreading, early-flowering tree. H 20ft (6m), S 15ft (5m). Has ovate, mid-green leaves. In early to mid-spring, bears large, cup-shaped, creamy-white flowers, 8–12in (20–30cm) across, violet-pink at the bases, with thick petals. Z6–8
M. Black Tulip ('Jurmag1') (illus. p.70). Vigorous, deciduous, densely branched tree. H to 40ft (12m), S 20ft (6m). Produces large, goblet-shaped, very dark wine-purple flowers, 8in (20cm) across or more, in early spring, before the large, ovate, mid-green leaves unfold. Z5–9
M. x brooklynenesis 'Yellow Bird' (illus. p.70). Variable, deciduous, upright, slightly pyramidal tree. H 40ft (12m), S 20ft (6m). Has ovate, mid-green leaves and deep

MAGNOLIA

yellow flowers, 4in (10cm) or more across, tinged green at the bases, in spring. Z3–8
M. 'Butterflies' (illus. p.70). Deciduous, upright tree. H and S 10ft (3m). Has ovate, mid-green leaves. Goblet- to tulip-shaped, bright deep yellow flowers, 4in (10cm) across or more, are borne in mid-spring opening widely as they mature to reveal orange-red stamens. Z3–8
M. 'Caerhays Belle'. Spreading, deciduous tree. H 30ft (10m), S 22ft (7m). Has ovate, mid-green leaves, tapering at both ends. Bears large, cup-shaped, light salmon-pink flowers, to 1ft (30cm) across, paler inside, in early spring before the leaves. Red fruiting cones, like small cucumbers, ripen in late summer. May be damaged by spring frosts. Z6–9
M. campbellii. Deciduous tree, upright when young, later spreading. H 50ft (15m), S 30ft (10m). Bears large, slightly fragrant, white, red, or pale to deep pink flowers, to 1ft (30cm) across, on leafless branches from late winter to mid-spring on trees 15–20 years old or more. Z7–10. **'Charles Raffill'** bears large, fragrant, cup-shaped, purplish-pink flowers on trees at least 15 years old. Leaves are large, ovate and mid-green. **'Darjeeling'** has large, very deep pink flowers. **subsp. mollicomata** (illus. p.70) has lilac-pink flowers slightly earlier in the year.
M. 'Daphne'. Compact, upright, deciduous shrub or small tree. H 15–22ft (5–7m), S 6–10ft (2–3m). Has broadly ovate, pointed, mid-green leaves. In late spring, bears deep yellow flowers, 4in (10cm) across; they are erect and held above the leaves at first, to avoid damaging frosts. Z5–10
M. 'Daybreak'. Floriferous, small to medium-sized, columnar tree. H 20–25ft (6–8m), S 6–10ft (2–3m). Has large, ovate, mid-green leaves. Large, upright, very fragrant, light rose-pink flowers, to 10in (25cm) across, deeper pink outside, are borne in late spring, with the emerging leaves; flowers from an early age. Z5–8
M. delavayi. Evergreen, rounded, dense shrub or tree. H and S 30ft (10m). Large, slightly fragrant, bowl-shaped, parchment-white flowers, to 8in (20cm) across are short-lived and open intermittently from midsummer to early fall. Large, ovate leaves are deep blue-green above and bluish-white beneath. Z7–10
M. denudata, syn. *M. heptapeta* (Lily tree, Yulan). Deciduous, spreading shrub or tree. H and S 30ft (10m). Bears profuse fragrant, cup-shaped, white flowers, to 6in (15cm) across, from mid- to late spring before ovate, mid-green leaves emerge. Z3–8
M. doltsopa, syn. *Michelia doltsopa* (illus. p.70). Evergreen, rounded tree. H 25–50ft (8–15m), S 15–30ft (5–10m). Has ovate, glossy, dark green leaves, paler beneath, and fragrant, bowl-shaped, white to pale yellow flowers, 3–4in (7–10cm) across, from spring to early summer. Z9–11
M. 'Elizabeth' (illus. p.70). Small to medium-sized, deciduous, multi-stemmed conical tree. H to 30ft (10m), S to 20ft (6m). In spring, produces cupped, fragrant, primrose-yellow flowers, to 6in (15cm) across, with red stamens. Ovate, dark green leaves, to 8in (20cm) long, are coppery in color when they unfold in spring. Z3–8